Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Small x Big y
Selected Domain	Domain 2: Orbyte Smart SME Challenge
Selected SDG Track (1–17)	12 (Responsible Consumption and Production), target 12.3. Also supports SDG 8 (targets 8.2 and 8.3). If the form allows one number only, 12 has the strongest measurable baseline (food waste).
Project / Solution Name	RestockAI
Name (Leader)	Eishvar A/L Rajesh Khanna
Email & Phone (Leader)	eishvar2020@gmail.com, 0175455259
University Affiliation(s)	AIMST University
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia's micro and small food businesses (kopitiams, cafés, campus eateries) decide every week what to buy for the kitchen. Most do it by hand, from memory, notebooks, WhatsApp chats and spreadsheets. We estimate that an owner spends about two days of calculation per purchasing cycle (a team estimate that we will validate with owners in Phase 2). The work is hard to do well: menus have dozens of dishes, each dish uses several ingredients, and demand moves with the weekday, university semesters, school holidays and festivals.

The mistakes are costly in two directions. Buy too much and fresh items such as milk, bread, fruit and chicken expire before they are used, which wastes money and food. Buy too little and popular dishes sell out, which loses sales and customers. Owners rarely have time to measure which mistake they make, so the same errors repeat.

This matters beyond one shop. The UN Environment Programme's Food Waste Index Report 2024 estimates that the world wasted 1.05 billion tonnes of food in 2022, and that food service (restaurants and caterers) caused 28% of it. Thin-margin SMEs feel every ringgit of spoiled stock, and the owner's time is the scarcest resource they have.

The people affected are owners, managers and kitchen staff of small food businesses, many with mixed digital confidence, who need a tool that is simple, explains itself, and leaves them in control. Their suppliers, who receive vague or late orders, are affected too.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained

from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

RestockAI is a purchase-planning module for a shared business hub such as Orbyte. The journey has five steps. The owner or manager picks how many days ahead to plan and presses Generate (trigger). The app combines the shop's sales history, recipes, current stock with expiry dates, suppliers and a calendar of holidays and events (connected data). It forecasts demand per dish, turns that into ingredient needs, subtracts usable stock, adds a safety buffer, and rounds up to whole supplier packs, buying perishables in several trips (decision logic). A manager submits the order and the owner approves or rejects it (human approval). The app then writes a WhatsApp-ready purchase note for each supplier, updates stock when goods arrive, and records every step in an activity log (visible outcome).

AI is used only where it adds value. The forecast and safety buffers are transparent statistics, not a language model: each buffer is measured from the app's own past forecast errors and shown on every purchase line with a "Why?" in plain words. On our synthetic demo data the daily-total forecast is off by about 13% on a typical day, against about 28% for a simple 4-week average. The language model is used in two places only: a Help chat that explains how to use the app (it sees no business data and changes nothing), and a reader for PDF sales reports, whose output a person checks and can fix in a preview before anything is saved. If AI is unavailable, manual forms and CSV or Excel upload still work.

From the SDC Phase 1 workshops we use GenAI APIs (through OpenRouter), database architecture (PostgreSQL with SQLAlchemy) and cloud deployment (Replit). All data in the prototype is synthetic.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

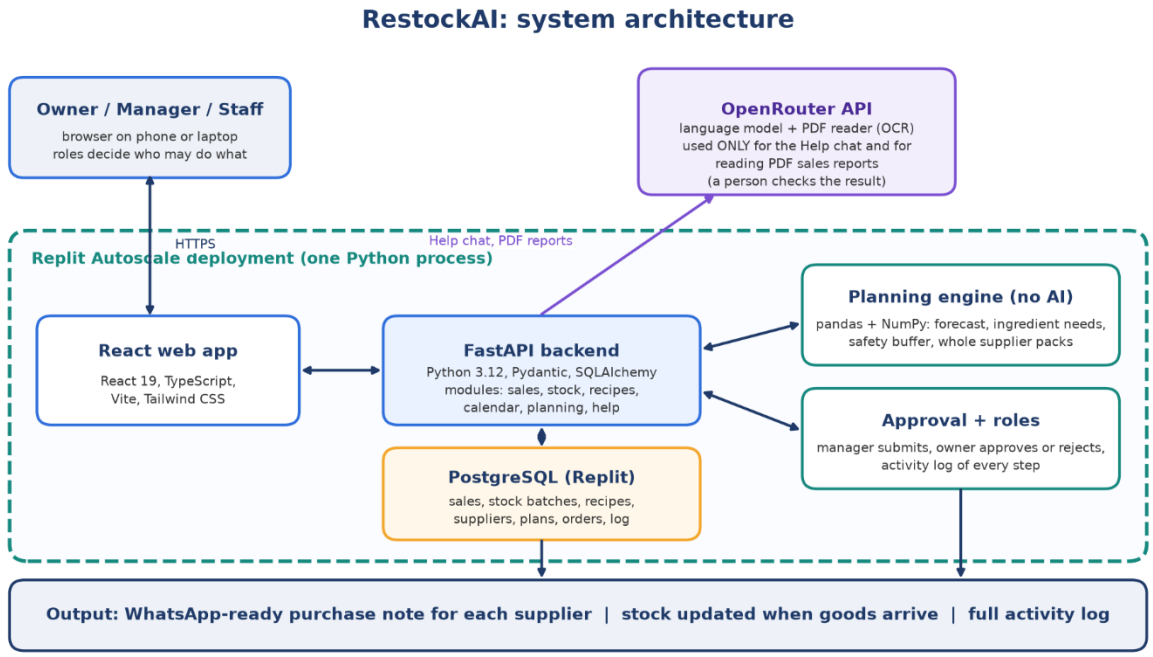
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

RestockAI is deployed as a single web application on Replit. One Python server answers the API requests and also serves the web pages.

- **Frontend:** React 19 and TypeScript, built with Vite, styled with Tailwind CSS, using TanStack Query for data and Recharts for charts. It is mobile-friendly and uses plain words, with three roles (owner, manager, staff).
- **Backend:** Python 3.12 with FastAPI, Pydantic for data checks, SQLAlchemy for database access, and Uvicorn as the server. Each feature is its own module: sales data, inventory, recipes, calendar, planning, settings, activity log and help.
- **Planning engine:** pandas and NumPy. It forecasts demand per dish using weekday, semester, holiday and event patterns. It turns that into ingredient needs, subtracts stock that hasn't expired, adds a safety buffer measured from past forecast errors, and rounds up to whole supplier packs. The engine uses no database and no AI. Automated tests compare it to a reference prototype.
- **Database:** PostgreSQL on Replit, with SQLite for local development. It holds sales, stock batches, recipes, suppliers, plans, orders and the activity log.
- **GenAI API:** a language model reached through the OpenRouter API (model set in configuration) **[YOU: write the exact model id]**. It is used in two places only:
    - the Help chat, which explains how to use the app and sees no business data;
    - reading PDF sales reports, using OpenRouter's PDF reader, which does the OCR. A person checks and fixes the result in a preview before anything is saved.

  If AI is unavailable, manual forms and CSV or Excel upload still work.

- **Output:** a WhatsApp-ready purchase note for each supplier, updated stock, and a full activity log. The app never pays or sends money.
- **Quality:** 289 automated tests (pytest) and 12 frontend helper tests.



Yes. We used Replit as the cloud platform: it hosts the deployed prototype (Autoscale deployment) and its managed PostgreSQL database.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define.We started from the Domain 2 brief: owners of micro and small businesses who work through WhatsApp, spreadsheets and paper, with mixed digital confidence. We chose one recurring burden, purchase planning, and wrote its steps down: forecast, ingredients, stock, order, approval. **[CHECK: this answer must not claim owner interviews unless they really happened. In Phase 2 we plan to test with real owners.

Ideate and prototype. Early on we decided that the maths would be transparent statistics and that AI would only help with messy input. We then built in small milestones, from the engine to the full journey, and kept every screen in plain words, with tooltips and empty states.

Test.We test technical viability with 289 automated tests, a comparison against a reference prototype, and a forecast audit on eight simulated restaurant types (12% to 22% error, close to the lowest in every type). We also documented honest limits, such as sold-out days hiding demand.

Deployment readiness. The app is multi-tenant, has roles (owner, manager, staff), requires human approval before any order, logs every action, never pays or transfers money, and already runs on Replit with PostgreSQL.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 12, Target 12.3: by 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains. Also **8.2** (higher productivity through technology and innovation) and 8.3 (support for growth of micro, small and medium-sized enterprises).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	(1) Food service caused 28% of the 1.05 billion tonnes of food waste in 2022 (about 294 million tonnes) (UNEP Food Waste Index Report 2024). (2) Manual purchase planning takes about 16 hours (two days) per cycle (team estimate, to be validated with pilot owners). (3) A simple 4-week average forecast is off by about 28% on a typical day on our synthetic data. The waste baseline for a single shop is measured in the app during Phase 2 onboarding.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Kilograms (and RM) of expired stock discarded per month, taken from the app's discard log; hours spent planning per cycle; daily-total forecast error (WAPE).

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	(1) At least 20% less expired stock discarded than the pilot shop's own previous four weeks. (2) Planning time cut from about 16 hours to under 1 hour (over 90%). (3) Daily-total forecast error of 15% or lower on pilot data (13% on synthetic demo data today). These are targets, not results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Name	**Type (code/dataset/model)**	**License**
FastAPI 0.159	code	MIT
Uvicorn 0.54	code	BSD-3-Clause
SQLAlchemy 2.1	code	MIT
Pydantic 2.14	code	MIT
pandas 3.0	code	BSD-3-Clause
NumPy 2.5	code	BSD-3-Clause
psycopg2-binary 2.9	code	LGPL with exceptions
python-dotenv 1.2	code	BSD-3-Clause
openpyxl 3.1	code	MIT
python-multipart 0.0.32	code	Apache-2.0

tzdata 2026.5	code / data	Apache-2.0
Anthropic Python SDK 2.1	code	Apache-2.0
OpenAI Python SDK 3.36 (used to call OpenRouter)	code	Apache-2.0
httpx 0.28 (testing)	code	BSD-3-Clause
pytest 9.1 (testing)	code	MIT
matplotlib (only in data/generate.py, draws one chart)	code	Matplotlib licence (PSF-style)
React and React DOM 19.3	code	MIT
React Router 7.18	code	MIT
TanStack Query 5.104	code	MIT
Recharts 3.10	code	MIT
Vite 8.3 and @vitejs/plugin-react 6.1	code	MIT
Tailwind CSS 4.3 and @tailwindcss/vite	code	MIT
TypeScript 6.0	code	Apache-2.0
oxlint 1.89 (code checker)	code	MIT
deepseek/deepseek-v4.1-flashvia OpenRouter (Help chat and PDF reading)	model	That model’s own licence and terms, plus OpenRouter’s terms
OpenRouter’s PDF reader (Mistral OCR engine)	model / service	OpenRouter and provider terms
Malaysian public holiday dates (data/calendar_my.csv)	dataset	Public dates, compiled by the team
Demo sales, menu, recipes, suppliers, stock, calendar (`data/`)	dataset	Synthetic, generated by the team’s own scripts; no personal data

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

25/10/26

[Date]